

		$E = (2.0)(2.0) + (2.0)(1.0) + (2.0)(1.0)$ Solving simultaneous equations: $E = 6.0 \text{ V}$
24	B	To shift the balance point to near Y, only increasing the resistance of the variable resistor R will reduce the pd across wire XY (potential divider) Reducing Q will shift the point to near X. Varying P has no effect as current flow is zero at balance. Likewise, replacing the wire with one of higher resistance per unit length will shift the balance point towards X.
25	C	Currents in X and Y are in the same direction. X and Y will experience